

RM05

## Blood Relations

2-3 questions in every RAS Prelims paper, without a single exception. The photograph type and the coded-symbol type are the two that repeat most. At full marks this is the cheapest block in the paper - a trained student clears all three in about 90 seconds.

A blood relation question is a family tree hidden inside two or three sentences. Draw the tree and the answer is obvious; try to hold it in your head and you will get it wrong. This chapter teaches you the drawing, not the facts.

### Blood Relations

#### Direct chain

- A is father of B, B is mother of C
- Read one link at a time
- Answer = relation of first to last

#### Photograph / pointing

- Pointing to a man, X said...
- Start from the SPEAKER
- 'only son', 'only daughter' clues
- Gender from he/she in the stem

#### Coded relations

- + - x / stand for relations
- Decode pair by pair
- First letter holds the relation
- 'Cannot be determined' when gender is open

#### Family-tree puzzles

- 5-7 members, A to F
- Married couple clue
- Daughter-in-law / widow clue
- Count males and females

#### Generation counting

- Parent step = +1 generation
- Sibling / spouse step = same level
- 2 steps = grandparent, 3 = great-grand

#### Maternal vs paternal

- Mama = mother's brother
- Chacha = father's brother
- Bua = father's sister
- Mausi = mother's sister
- Nana vs Dada

### What the question looks like, and where the marks leak

A blood relation question gives you a small chain of family facts and asks how the first person is related to the last. Sometimes it is a plain chain - 'A is the father of B, B is the mother of C'. Sometimes it is a person pointing at a

photograph. Sometimes the relations are hidden behind symbols like + and x. The underlying job is always the same: build a family tree.

- Direct chain puzzles - two to four sentences of relations.
- Pointing / photograph type - one long sentence spoken by a person.
- Coded relations - '+', '-', 'x', '/' each stand for a relation.
- Family-tree puzzles with five to seven members named A to G.
- Generation counting - how many generations does this family span.
- Maternal or paternal identification - uncle, aunt, grandfather.

Marks leak in exactly three places. One: the student reasons verbally instead of drawing, and loses the thread at the third link. Two: the student ignores gender and calls a maternal uncle a paternal uncle. Three: the student reverses the direction of a relation - 'X is the son of Y' gets drawn as Y below X. Every one of those three is a drawing discipline problem, not an intelligence problem.

#### TRAP

Never answer a blood relation question without putting pen to paper. The examiner writes the sentences in a deliberately tangled order precisely so that verbal reasoning fails. Ten seconds of drawing beats forty seconds of thinking.

#### ASKED IN RAS

Blood relations has appeared in every RAS Prelims paper. The photograph form came in 2023 ('His mother's only daughter is my mother' - answer: maternal uncle) and in 2016 ('the son of the only son of my grandmother' - answer: brother). The coded-symbol form came in 2021, the in-law form in 2024 and the plain chain in 2018 and 2013. Prepare all four; none of them skips a year for long.

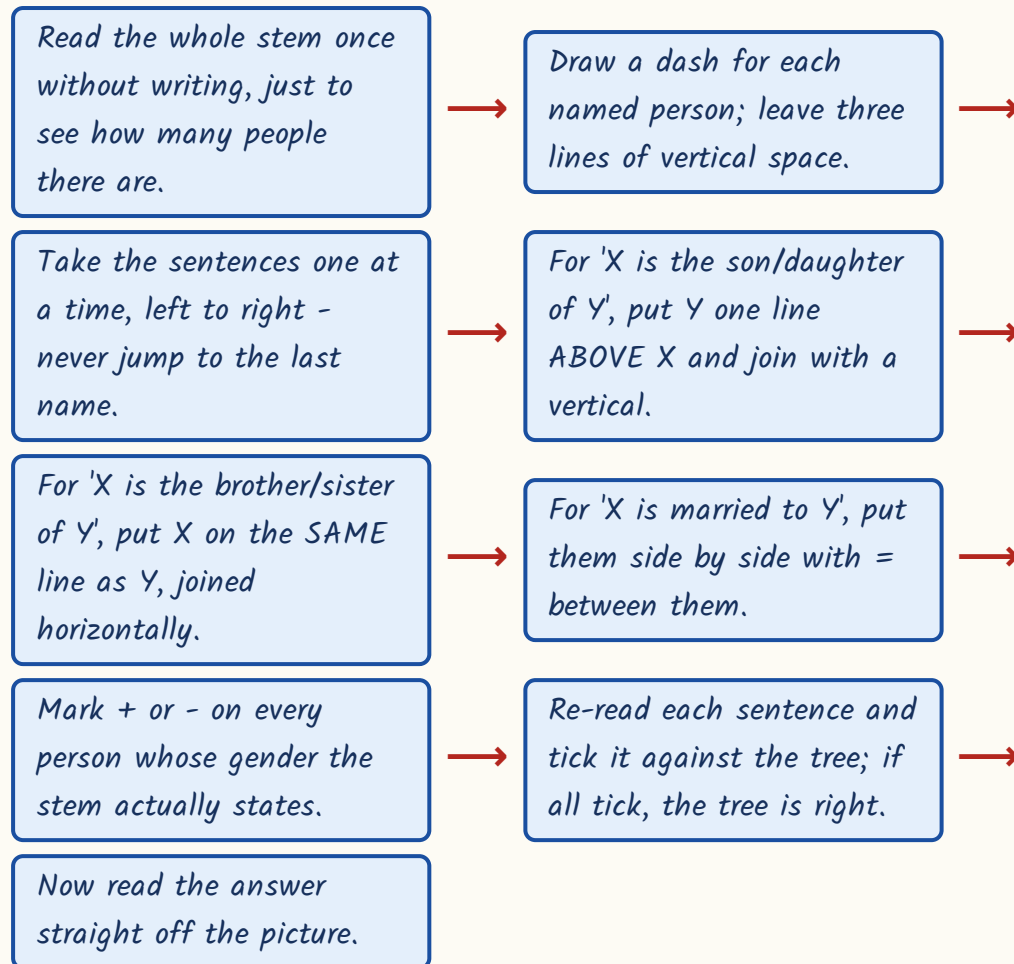
## *The notation - draw the tree the same way every single time*

*Use one fixed set of marks. If you invent notation afresh in the exam hall you will misread your own diagram. Learn these five marks now and never change them.*

- *+ written next to a name means that person is MALE.*
- *- written next to a name means that person is FEMALE.*
- *= or a double horizontal line between two names means they are a married couple.*
- *A vertical line downward means parent to child.*
- *A single horizontal line means siblings, standing on the same level.*

*The rule that keeps the diagram honest is the level rule. The eldest generation sits on the top line. Their children sit on the next line down. Grandchildren on the line below that. Two people on the same line are always of the same generation - siblings, cousins, or a husband and wife. Never let a parent and a child share a line.*

### The method - follow these steps with pen on paper



### Two habits that fix half the errors

- Solve the 'only son / only daughter / only child' clue first - it forces a unique person.
- A married couple in an exam puzzle are always the parents of the same children.
- The spouse of a parent is therefore also a parent - use this without hesitation.
- Label the NODE, not the sentence: one person can be a son and a father at once.
- Gender that is never stated stays blank. Do not silently assume a man.

**YAAD RAKHO**

Yaad rakho - PLUS is PURUSH (male), MINUS is MAHILA (female). Two letters, two symbols, and you will never mix them up under time pressure.

## The relationship dictionary - and the maternal-paternal split

RPSC does not ask 'who is the uncle'. It asks 'who is the maternal uncle'. The option list will always contain both the maternal and the paternal version, so knowing which side of the family you are on is worth the whole mark. Maternal means through the mother. Paternal means through the father. That is the entire rule; the vocabulary below just applies it.

### Blood relatives - English and the common Hindi term

| English                     | Hindi             | Exactly who                           |
|-----------------------------|-------------------|---------------------------------------|
| Maternal uncle              | Mama              | Mother's brother                      |
| Maternal aunt (by marriage) | Mami              | Wife of mother's brother              |
| Maternal aunt (by blood)    | Mausi             | Mother's sister                       |
| Paternal uncle              | Chacha            | Father's brother                      |
| Paternal aunt (by marriage) | Chachi            | Wife of father's brother              |
| Paternal aunt (by blood)    | Bua               | Father's sister                       |
| Paternal grandfather        | Dada              | Father's father                       |
| Maternal grandfather        | Nana              | Mother's father                       |
| Cousin                      | -                 | Child of any uncle or aunt            |
| Nephew / niece              | Bhatija / Bhatiji | Son / daughter of a brother or sister |

### Which side am I on - the only distinction RPSC tests

| <b>MATERNAL side (through the mother)</b>     | <b>PATERNAL side (through the father)</b>     |
|-----------------------------------------------|-----------------------------------------------|
| Mother's brother = maternal uncle (mama)      | Father's brother = paternal uncle (chacha)    |
| Mother's sister = maternal aunt (mausi)       | Father's sister = paternal aunt (bua)         |
| Wife of mother's brother = mami               | Wife of father's brother = chachi             |
| Mother's father = maternal grandfather (nana) | Father's father = paternal grandfather (dada) |
| Mother's mother = maternal grandmother (nani) | Father's mother = paternal grandmother (dadi) |
| Reached by going UP through a female parent   | Reached by going UP through a male parent     |

### In-laws - three rules cover every case

| <b>Relation to you</b>        | <b>Who they are</b>                            |
|-------------------------------|------------------------------------------------|
| Father-in-law / mother-in-law | The parent of your spouse                      |
| Son-in-law / daughter-in-law  | The spouse of your child                       |
| Brother-in-law                | Your spouse's brother OR your sister's husband |
| Sister-in-law                 | Your spouse's sister OR your brother's wife    |

#### TRAP

Do not confuse a cousin with a nephew. The child of your uncle or aunt is your COUSIN - same generation, same line on the tree. The child of your brother or sister is your NEPHEW or NIECE - one line below you. Draw the level and the confusion disappears.

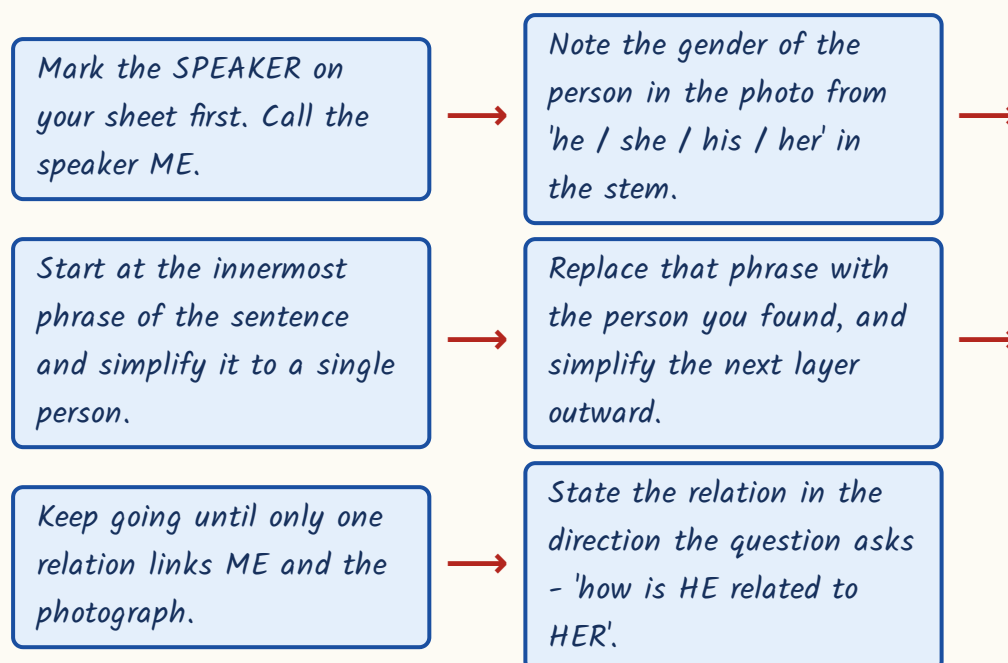
#### ASKED IN RAS

Asked in RAS Pre 2024 - A and B are a married couple, C and D are their son and daughter, E is married to D. How is E related to B? Answer: son-in-law. The in-law form is now a standing favourite; learn the three in-law rules above by heart.

## Worked examples – the pointing and photograph type

In this type one person speaks a single sentence about someone in a photograph. The whole difficulty is that the sentence has to be unwound from the inside out. There is a fixed method for it, and once you follow the method these become the fastest questions in the section.

### Photograph type – unwinding the sentence



Worked example 1 (asked in RAS Pre 2023). Pointing to a man in a photograph, Reena said, 'His mother's only daughter is my mother.' How is the man related to Reena?

- Mark Reena as ME on the sheet. The photo person is a MAN, so mark him +.
- Innermost phrase: 'his mother's only daughter'. The man's mother has one daughter.
- The man is her son, so that only daughter is the man's own SISTER.
- The sentence now reads: the man's sister is my mother.

- So Reena's mother and the man are brother and sister - same line on the tree.
- The man is therefore the brother of Reena's mother.
- Brother of the mother = MAMA = maternal uncle. Answer: maternal uncle.

Worked example 2. Pointing to a boy, a woman said, 'He is the son of the only brother of my mother's daughter.' The woman has only one brother. How is the woman related to the boy?

- Mark the woman as ME, female, and the boy as +.
- Innermost phrase: 'my mother's daughter'. That is the speaker herself, the woman.
- The sentence becomes: he is the son of the only brother of ME.
- The woman has only one brother; that brother is the boy's father.
- So the boy's father and the woman are brother and sister on the same line.
- The question asks how the WOMAN is related to the BOY - reverse the direction.
- Sister of the boy's father = BUA = paternal aunt. Answer: paternal aunt.

#### TRAP

The words 'other than myself' and 'only' are not filler. 'My mother's only son' is the speaker himself if the speaker is male; add 'other than myself' and it becomes his brother. Read those four words twice - they flip the answer completely.

### Coded relations - when symbols replace the words

Here the paper first defines a code - 'A + B means A is the father of B', and so on - then gives you an expression like 'K / L x M' and asks how K is related to M. Students lose these by trying to read the expression as one sentence. Do



not. Break it into overlapping pairs and decode each pair separately onto the same tree.

- For ' $P + Q - R$ ', decode  $P+Q$  first, then  $Q-R$ . The pairs overlap at  $Q$ .
- In every pair, the FIRST letter holds the stated relation to the second.
- So in ' $P + Q$ ' with  $+$  meaning father,  $P$  is the father, not  $Q$ .
- Draw each decoded pair onto a single growing tree, not three separate scraps.
- If the question asks 'which expression means  $X$  is the ... of  $Y$ ', build a small tree for EACH option separately. It is faster than arguing verbally.
- If the answer depends on the gender of a middle letter that was never stated, the correct choice is 'cannot be determined'.

Worked example 3 (asked in RAS Pre 2021). Code: ' $A + B$ ' =  $A$  is the father of  $B$ ; ' $A - B$ ' =  $A$  is the sister of  $B$ ; ' $A \times B$ ' =  $A$  is the mother of  $B$ ; ' $A / B$ ' =  $A$  is the brother of  $B$ . How is  $K$  related to  $M$  in ' $K / L \times M$ '?

- Split into pairs:  $K / L$ , then  $L \times M$ .
- $K / L$  uses ' $/$ ', so  $K$  is the BROTHER of  $L$ . Put  $K$  and  $L$  on the same line; mark  $K$  as  $+$ .
- $L \times M$  uses ' $\times$ ', so  $L$  is the MOTHER of  $M$ . Put  $M$  one line below  $L$ , vertical line.
- Now read the tree.  $K$  is on  $L$ 's line;  $M$  is  $L$ 's child.
- $K$  is therefore the brother of  $M$ 's mother.
- Brother of the mother = maternal uncle. Answer: maternal uncle.

Worked example 4. Same style of code: ' $A + B$ ' =  $A$  is the father of  $B$ ; ' $A - B$ ' =  $A$  is the sister of  $B$ ; ' $A \times B$ ' =  $A$  is the mother of  $B$ ; ' $A / B$ ' =  $A$  is the brother of  $B$ . Which expression means ' $P$  is the paternal grandfather of  $Q$ '?

- Paternal grandfather means: father of the FATHER. Two male parent steps upward.

- So you need  $P$  to be the father of some middle person  $R$ , and  $R$  to be the father of  $Q$ .
- ' $P + R$ ' gives  $P$  is the father of  $R$ . Correct for the first step.
- ' $R + Q$ ' gives  $R$  is the father of  $Q$ . Correct for the second step.
- Joining them:  $P + R + Q$ . Answer:  $P + R + Q$ .
- Check the wrong options - ' $P \times R + Q$ ' makes  $P$  the MOTHER of  $R$ , so  $P$  would be the paternal grandmother, not grandfather.

**TRAP**

The commonest error in coded relations is reversing the pair. ' $A \times B$ ' where  $\times$  means mother makes  $A$  the mother of  $B$  -  $A$  is above  $B$  on the tree.

Students who write  $B$  above  $A$  get a consistent-looking but exactly inverted answer, and the paper always puts that inversion in the options.

## Family-tree puzzles and generation counting

The puzzle form names five to seven members -  $A, B, C, D, E, F$  - and scatters the clues out of order. The trick is clue selection. Start with the clue that pins a person down completely: a married couple, an 'only son', a 'daughter-in-law of', or a stated gender. Weak clues like ' $C$  is the brother of  $A$ ' go in later, once there is a frame to hang them on.

Worked example 5. In a family of six members  $A, B, C, D, E$  and  $F$ :  $A$  and  $B$  are a married couple,  $A$  being the male member.  $D$  is the only son of  $C$ , who is the brother of  $A$ .  $E$  is the sister of  $D$ .  $B$  is the daughter-in-law of  $F$ , whose husband has died. How is  $E$  related to  $A$ , and how is  $F$  related to  $D$ ?

- Strongest clue first:  $A$  and  $B$  are a couple,  $A$  is male. Write  $A+ = B-$  on one line.

- Next: B is the daughter-in-law of F. So F is a parent of B's husband, that is a parent of A.
- F's husband has died, so F is the surviving parent - F is A's MOTHER. Put F- one line above A.
- C is the brother of A, so C sits on A's line, joined horizontally, and F is C's mother too.
- D is the only son of C - put D+ one line below C. E is D's sister, so E- goes beside D, also C's child.
- Now read off. E is the daughter of C, and C is A's brother. So E is A's NIECE.
- F is the mother of C, and D is C's son. Two parent steps above D, so F is D's GRANDMOTHER.

### Counting generations

- Every 'father of / mother of / son of / daughter of' step changes the generation by one.
- Every 'brother of / sister of / husband of / wife of' step keeps the same generation.
- Two parent-steps up = grandparent. Three parent-steps up = great-grandparent.
- To count how many generations a family spans, count the distinct LINES on your tree.
- Count the steps on paper. Never guess from the name of the relation.

Worked example 6. A is the mother of B. B is the brother of C. C is the mother of D. D is the brother of E. How many generations are there in this family?

- A is the mother of B - that is one parent step. A on line 1, B on line 2.
- B is the brother of C - sibling step, no change. C also sits on line 2.

- *C is the mother of D - parent step. D goes on line 3.*
- *D is the brother of E - sibling step. E also sits on line 3.*
- *Distinct lines used: line 1 (A), line 2 (B and C), line 3 (D and E).*
- *Answer: three generations. Note that the chain has four sentences but only two parent steps.*

## **The 60-second checklist**

- *Draw before you think. Dashes for people, levels for generations.*
- *Fix the 'only child' clue and the married-couple clue first.*
- *Mark + and - only where the stem actually gives gender.*
- *Read chains and coded expressions pair by pair, left to right.*
- *For photograph questions, mark the speaker as ME and unwind from the inside.*
- *Before choosing, check maternal versus paternal - the wrong side is always an option.*
- *Check the direction the question asks: 'how is X related to Y', not Y to X.*
- *For a 'how many male members' question, count each named person once; a married-in spouse is a family member but not a blood relation.*
- *Tick every sentence of the stem against the finished tree.*

### **ASKED IN RAS**

*Asked in RAS Pre 2018 - P is the brother of Q, Q is the son of R, R is the wife of S; how is P related to S? Answer: son. The chain type looks trivial and is still worth full marks in twenty seconds, so never skip it looking for something harder.*

**YAAD RAKHO**

*Yaad rakho - MAMA has an M like MOTHER, so mama is the mother's brother. CHACHA does not, so chacha belongs to the father's side. One letter, and the maternal-paternal trap is closed.*

**REVISION RECAP**

1. Every blood relation question is a family tree; draw it, never reason verbally.
2. Notation: + male, - female, = married couple, vertical line parent-to-child, horizontal line siblings.
3. Eldest generation on the top line; never put a parent and a child on the same line.
4. Read chains one link at a time, left to right; never jump from the first name to the last.
5. Sibling, husband and wife steps keep the generation; parent and child steps change it by one.
6. Solve 'only son', 'only daughter' and 'only child' clues first - they force a unique person.
7. A married couple in an exam puzzle are the parents of the same children.
8. Mama = mother's brother; chacha = father's brother; bua = father's sister; mausi = mother's sister.
9. Mami is the wife of the mama; chachi is the wife of the chacha.
10. Nana is the maternal grandfather; dada is the paternal grandfather.
11. Child of an uncle or aunt = cousin; child of a brother or sister = nephew or niece.
12. In-laws: parent of spouse, spouse of child, and sibling of spouse or spouse of sibling.
13. Photograph type: mark the speaker as ME, take the gender from he/she in the stem, unwind from the inside.
14. Coded relations: decode overlapping pairs; the first letter of each pair holds the relation.
15. For 'which expression means...', build a separate small tree for every option.
16. If the answer turns on a gender the stem never states, choose 'cannot be determined'.
17. Two parent-steps up is a grandparent; three is a great-grandparent - count, do not guess.
18. Finish by re-reading every sentence against the tree; if all of them tick, the tree is right.